Math 4141: Geometry and Differential Calculus

Dr. Md Tusher Mollah

Assistant Professor, Department of Natural Sciences
Islamic University of Technology, Bangladesh
PhD and Postdoc from the Department of Civil and Mechanical Engineering

Math 4141: Geometry and Differential Calculus — Credit: 4.00

Course Contents:

2D Coordinate Geometry: Change of axes: Transform of coordinates, Simplification of equations of curves; Pair of straight lines: Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, Angle between the lines, Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.

3D Coordinate Geometry: Rectangular coordinates, Direction cosines and angle between two lines, The plane and the straight lines, The equation of a sphere, The standard forms of equations of the central conicoid, cones and cylinders.

Differential Calculus: Limits, Continuity and Differentiability; Differentiation of explicit and implicit functions and parametric equations; Significance of derivatives, Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem; Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders; Expansion of functions by differentiation and integration; Partial differentiation; Euler's Theorem; Tangent, maximum and minimum values of functions and points of inflection; Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule; Curvature, the center of curvature and chord of curvature; Evolutes and involutes; Asymptotes, Envelopes, Curve tracing.

Exercise

Q. Find the value of λ if the following equations may represent pairs of straight lines $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$.

Solution: Given $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$

Now compare with equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then we get

$a = 12, h = -5, b = 2, g = \frac{11}{2}, f = -\frac{5}{2}$ and $c = \lambda$

We have $\Delta = 0$

Find the value of λ yourself.

Q. Find the value of λ if the following equations may represent pairs of straight lines $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$.

Exercise

Q. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represent a pair of parallel straight lines, then prove that $h^2 = ab$ and $bg^2 - af^2 = 0$.

Solution: Given $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b}$$

Since pair of straight lines are parallel, we have $\theta = 0$ then $h^2 = ab$
(Expand the calculations yourself.)

We know that the condition of a pair of straight line is:

$$\Delta = 0$$

$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow h^2c + 2fg\sqrt{ab} - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow -bg^2 + 2fg\sqrt{a}\sqrt{b} - af^2 = 0$$

$$\Rightarrow \left(\sqrt{b}\right)^2 g^2 - 2\sqrt{b}gf\sqrt{a} + (\sqrt{a})^2 f^2 = 0$$

$$\Rightarrow (\sqrt{b}g)^2 - 2\sqrt{b}gf\sqrt{a} + (\sqrt{a}f)^2 = 0$$

$$\Rightarrow (\sqrt{b}g - \sqrt{a}f)^2 = 0$$

$$\Rightarrow \sqrt{b}g - \sqrt{a}f = 0$$

$$\Rightarrow \sqrt{b}g = \sqrt{a}f$$

$$\Rightarrow (\sqrt{b}g)^2 = (\sqrt{a}f)^2$$

$$\Rightarrow bg^2 = af^2$$

$$\therefore bg^2 - af^2 = 0$$

Reduce the equation in standard form

Q. Reduce the equation in standard form $9x^2 - 6xy + y^2 - 14x - 2y + 12 = 0$. Hence identify it.

Solution: Given that,

$$9x^2 - 6xy + y^2 - 14x - 2y + 12 = 0$$

Now compare with the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, we get

$a = 9, h = -3, b = 1, g = -7, f = -1$ and $c = 12$

We have $\Delta =$ *"Find the value yourself"* $-400 \neq 0$

and $ab - h^2 = 0$. So given equation represent Parabola.

Now,

$9x^2 - 6xy + y^2 - 14x - 2y + 12 = 0$

$=> 9x^2 - 6xy + y^2 = 14x + 2y - 12$

$=> (3x - y)^2 = 14x + 2y - 12$

To choose coordinates along perpendicular directions, add a constant (k) inside the square.

$=> (3x - y + k)^2 = 14x + 2y - 12 + k^2 + 6xk - 2ky$

$=> (3x - y + k)^2 = x(14 + 6k) + y(2 - 2k) + k^2 - 12$.....(1)

Reduce the equation in standard form – Solution Continues

Let $3x - y + k = 0$ and $x(14 + 6k) + y(2 - 2k) + k^2 - 12 = 0$ are mutually perpendicular each other-

$3.(14 + 6k) + (-1).(2k - 2) = 0$

$\therefore k = -2$

From equation (1) we get

$$(3x - y - 2)^2 = x(14 + 6.-2) + y(2 - 2.-2) + (-2)^2 - 12$$

$$=> (3x - y - 2)^2 = 2x + 6y - 8$$

$$=> \left(\frac{3x - y - 2}{\sqrt{3^2 + 1^2}}\right)^2 .10 = \frac{2x + 6y - 8}{\sqrt{2^2 + 6^2}} .\sqrt{40}$$

Let $X = \frac{3x-y-2}{\sqrt{10}}$ and $Y = \frac{2x+6y-8}{\sqrt{40}}$

$X^2 = Y.\sqrt{40}$

$$=> X^2 = \frac{\sqrt{40}}{10} .Y$$

$$=> X^2 = 4.\frac{\sqrt{40}}{40} .Y$$

$=> X^2 = 4.\frac{1}{\sqrt{40}} .Y$ Here $a = \frac{1}{\sqrt{40}}$

Reduce the equation in standard form

Q. Prove that $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$ represents a parabola. Also find the focus and directrix.

Solution

Given that

$$9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$$

Compare the given equation with the general second-degree equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0.$$

We obtain

$$a = 9, h = -12, b = 16, g = -10, f = -\frac{15}{2}, c = -60.$$

First,

$$h^2 - ab = (-12)^2 - (9)(16) = 144 - 144 = 0.$$

The determinant of the conic is

$$\Delta = \begin{vmatrix} 9 & -12 & -10 \\ -12 & 16 & -\frac{15}{2} \\ -10 & -\frac{15}{2} & -60 \end{vmatrix} = -\frac{15625}{4} \neq 0.$$

Thus, the conic is a parabola.

Now rearrange the equation:

$$9x^2 - 24xy + 16y^2 = 20x + 15y + 60,$$
$$(3x - 4y)^2 = 5(4x + 3y + 12).$$

The lines $3x - 4y = 0$ and $4x + 3y + 12 = 0$ are perpendicular.

Their slopes are $\frac{3}{4}$ and $-\frac{4}{3}$, and their product is -1.

Introduce coordinates along these perpendicular directions:

$$Y = \frac{3x - 4y}{\sqrt{3^2 + (-4)^2}} = \frac{3x - 4y}{5}, \qquad X = \frac{4x + 3y + 12}{\sqrt{4^2 + 3^2}} = \frac{4x + 3y + 12}{5}.$$

Since $3x - 4y = 5Y$ and $4x + 3y + 12 = 5X$, substitution gives

$$(5Y)^2 = 5(5X) \implies Y^2 = X.$$

Compare this with the standard form $Y^2 = 4aX$. Therefore,

$$4a = 1 \implies a = \frac{1}{4}.$$

The parabola opens in the positive X-direction.

Reduce the equation in standard form – Solution Continues

Q. Prove that $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$ represents a parabola. Also find the focus and directrix.

Focus

For $Y^2 = 4aX$, the focus in the (X, Y) coordinate system is $(a, 0)$. Here it is $\left(\frac{1}{4}, 0\right)$. Using the coordinate definitions,

$$\frac{3x - 4y}{5} = 0, \qquad \frac{4x + 3y + 12}{5} = \frac{1}{4}.$$

Thus,

$$3x - 4y = 0, \qquad 4x + 3y = -\frac{43}{4}.$$

Solving these simultaneous equations gives

$$\boxed{\text{Focus} = \left(-\frac{43}{25}, -\frac{129}{100}\right)}.$$

Directrix

For $Y^2 = 4aX$, the directrix is $X = -a$. Therefore $X = -\frac{1}{4}$, so

$$\frac{4x + 3y + 12}{5} = -\frac{1}{4},$$
$$4x + 3y + 12 = -\frac{5}{4}.$$

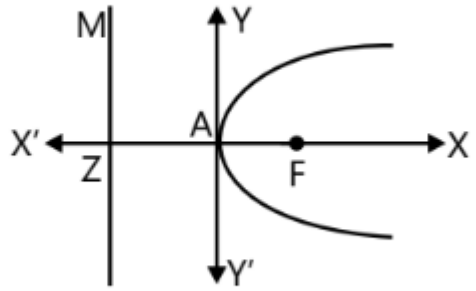
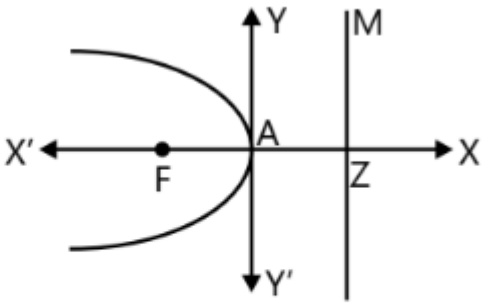
Therefore, the directrix is

$$\boxed{16x + 12y + 53 = 0}.$$

Hence, the given equation represents a non-degenerate parabola with focus $\left(-\frac{43}{25}, -\frac{129}{100}\right)$ and directrix $16x + 12y + 53 = 0$.

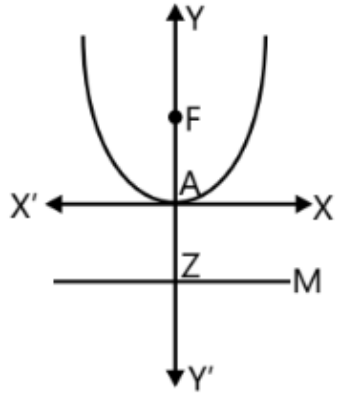
Supporting material

3. STANDARD FORMS OF PARABOLA

Right handed parabola: The equation of this type of parabola is of the form $y^2 = 4ax$, $a > 0$. See Fig. 10.7	 $y^2 = 4ax$ **Figure 10.7**
Left handed parabola: The equation of this type of parabola is of the form $y^2 = -4ax$, $a > 0$. See Fig. 10.8	 $y^2 = -4ax$ **Figure 10.8**

Supporting material

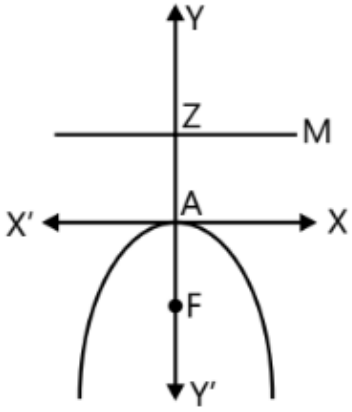
Upward parabola: The equation of this type of parabola is of the form $x^2 = 4ay$, $a > 0$. See Fig. 10.9



$x^2 = 4ay$

Figure 10.9

Downward parabola: The equation of this type of parabola is of the form $x^2 = -4ay$, $a > 0$. See Fig. 10.10



$x^2 = -4ay$

Figure 10.10

Supporting material

Equation of the parabola Properties	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Vertex (Co-ordinates)	(0, 0)	(0, 0)	(0, 0)	(0, 0)
Focus (Co-ordinates)	(a, 0)	(–a, 0)	(0, a)	(0, –a)
Latus rectum (length)	4a	4a	4a	4a
Axis (Equation)	y = 0	y = 0	x = 0	x = 0
Directrix (Equation)	x = –a	x = a	y = –a	y = a
Symmetry (about)	x-axis	x-axis	y-axis	y-axis

Thanks for your attention